

2(a)

any **four** from:

- 1 globular protein ;
 A spherical / ball-shaped
 I circular
- 2 acts as a catalyst ;
- 3, 4 further detail ; ;
 e.g. increases the rate of a reaction (cf. to no enzyme)
 A speeds up a reaction / makes the reaction go faster
 does not get used up (in the reaction)
 I does not take part in the reaction
 A can be re-used
 A is not changed (by the reaction)
 does not alter the (chemical) equilibrium (between reactants and products)
- 5 lowers the, activation energy / energy needed for substrate to reach transition state ;
- 6 detail ;
 e.g. holds substrate to place strain on bond (for bond to break)
 holds substrates, in position / close, for bonds to form
 facilitates transfer of, electrons / protons
- 7 effective in tiny quantities / AW ;
- 8 shows specificity / active site shape complementary to substrate / forms enzyme–substrate complex / lock and key or induced fit mechanism of action ;
- 9 AVP ;
 e.g. may need, cofactor / coenzyme / prosthetic group, to function
 each has an optimum, temperature / pH

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Question	Answer	Marks
2(b)	$\begin{array}{ccc} \text{carbon dioxide / CO}_2 & & \\ + & & \\ \text{water / H}_2\text{O} & & \end{array} \left. \vphantom{\begin{array}{c} \text{carbon dioxide / CO}_2 \\ + \\ \text{water / H}_2\text{O} \end{array}} \right\} ; \begin{array}{c} \xrightarrow{\text{body tissues}} \\ \xleftarrow{\text{lungs}} \end{array} \text{carbonic acid / H}_2\text{CO}_3 ;$ <i>one mark for each side of the equation</i> <i>allow one mark only if, wrong way round / carbonic acid shown as proceeding to dissociate</i>	2
2(c)	<i>any four from:</i> <i>primary:</i> sequence of amino acids not shown ; A cannot see primary structure of amino acids <i>secondary:</i> alpha / α, -helices shown ; beta / β, -pleated sheets shown ; (areas of) random arrangement shown ; <i>tertiary:</i> folding / coiling, (of polypeptide chain) shown / ref. to 3-D configuration ; globular shape shown ; interactions between, R-groups / side-chains, (of amino acids) not shown ; ref. to any two bond types not shown ; e.g. H bonds / disulfide bridges / ionic bonds / hydrophobic interactions folding allows presence of zinc / AW ;	4